Spurious regression and cointegration

Topic 9

GOALS

- To understand the **spurious regression** problem with trending (deterministic and stochastic) variables.
- To define **cointegration** and the cointegrating vector, and interpret them as a long-run equilibrium relationship.
- To appreciate the **superconsistency** of OLS estimators of cointegrating relationships.
- To test for cointegration via the Engle–Granger two-step procedure.

KEY CONCEPT

A vector of $\mathrm{I}(1)$ series $\mathbf{y}_t$ is **cointegrated** if some linear combination $\mathbf{a}'\mathbf{y}_t$ is $\mathrm{I}(0)$.

This topic is adapted from the non-stationary multivariate slide pack (Part II). $\mathrm{I}(d)$ denotes a process integrated of order d; $O_p(\cdot)$ is the usual stochastic-order notation.

---

1 The spurious regression problem

The phenomenon of finding a nonsensical relationship between two or more trending variables — simply because each is growing over time — is known as the **spurious regression problem**.

1.1 Deterministic trend

Suppose two *unrelated* variables Y_t and X_t are each generated by deterministic linear trends:

$$Y_t = \beta_1 + \beta_2 t + u_t, \qquad X_t = \alpha_1 + \alpha_2 t + v_t.$$

If Y_t is regressed on X_t, the common dependence on t makes the variables co-move, so the regression yields **spuriously significant** coefficients in large samples. This was recognised early in econometrics. A popular response was to **detrend** each variable (regress on t and save residuals $\tilde{Y}_t, \tilde{X}_t$) before regressing one on the other. It was eventually pointed out (Frisch & Waugh, 1933) that one need not detrend separately: including a time trend in the multiple regression of Y_t on X_t yields identical results — the origin of the Frisch–Waugh–Lovell theorem (Lovell, 1963).

1.2 Stochastic trend

The problem is far more severe with **stochastic** trends. Consider two independent random walks

$$Y_t = Y_{t-1} + \varepsilon_{1,t}, \qquad X_t = X_{t-1} + \varepsilon_{2,t},$$

with $\varepsilon_{1,t}, \varepsilon_{2,t}$ mutually independent. Regressing $Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t$:

The Granger–Newbold (1974) experiment 01

In a landmark Monte Carlo study, Granger and Newbold generated 100 pairs of independent random walks ($n = 50$) and regressed one on the other. **77 of the 100** regressions found a spuriously significant slope at the 5% level (where one would expect about 5). The Type I error rate was inflated roughly 15-fold. The effect is most dramatic for pure random walks (unit-root coefficient $= 1$), and is fundamentally about ignoring the strong autocorrelation in the regressors.
The experiments revealed that **standard regression theory does not apply** to non-stationary time series. Dickey–Fuller (1979) and related work showed that the relevant asymptotic distributions are non-standard.

`Regressing independent random walks on each other produces significance where there is no relationship.`

ℹ Algebra — regressions among unrelated unit-root processes with drift

Take two unrelated unit-root processes with non-zero drifts,

$$X_t = \alpha_X + X_{t-1} + \varepsilon_{1,t}, \qquad Y_t = \alpha_Y + Y_{t-1} + \varepsilon_{2,t},$$

with $X_0, Y_0 = O_p(1)$ and independent. Using the "big-bang" representation $Y_t = \alpha_Y t + \sum_{k=1}^{t} \varepsilon_{2,k} + Y_0$ (and similarly for X_t) together with the schoolboy-Gauss sums

$$\sum_{t=1}^{T} t = \frac{T(T+1)}{2} = O(T^2) \left(\text{so } \tfrac{1}{T^2} \textstyle\sum t \to \tfrac{1}{2}\right), \qquad \textstyle\sum_{t=1}^{T} t^2 = \frac{T(T+1)(2T+1)}{6} = O(T^3) \left(\text{so } \tfrac{1}{T^3} \textstyle\sum t^2 \to \tfrac{1}{3}\right),$$

the no-intercept OLS estimator $\hat{\beta}_{OLS} = \sum X_t Y_t / \sum X_t^2$ converges to the **ratio of the drifts**:

$$\hat{\beta}_{OLS} \xrightarrow{p} \frac{\alpha_Y}{\alpha_X} \quad \text{as } T \to \infty,$$

not to zero — despite the true data-generating process having no relationship between Y and X. This is the spurious regression result in algebraic form.

2 Cointegration

The spurious regression problem suggests avoiding levels of $\mathrm{I}(1)$ variables in regressions. But always differencing limits the economic questions we can ask (long-run relationships, equilibrium concepts). The key exception is **cointegration**, where two or more $\mathrm{I}(1)$ series may be regressed in levels *without* spurious results.

In general, a linear combination of non-stationary series is non-stationary, and the order of integration of a combination is that of its most-integrated component: $\mathrm{I}(1) + \mathrm{I}(0) \to \mathrm{I}(1)$; $\mathrm{I}(1) + \mathrm{I}(1) \to \mathrm{I}(1)$; $\mathrm{I}(1) + \mathrm{I}(2) \to \mathrm{I}(2)$. Cointegration is the special case where the combination is *less* integrated than its parts.

What is cointegration? 02

Definition. Suppose $\mathbf{y}_t = (y_{1t}, \dots, y_{Nt})'$ is an N-dimensional vector time series with $\{y_{it}\} \sim \mathrm{I}(1)$ for every i. We say $\mathbf{y}_t$ is **cointegrated with cointegrating vector $\mathbf{a}$** if there exists a vector of finite real constants $\mathbf{a} = (a_1, \dots, a_N)' \neq \mathbf{0}$ such that

$$\{\mathbf{a}'\mathbf{y}_t\} \sim \mathrm{I}(0), \qquad t \in \mathbb{Z}. \qquad \square$$

The linear combination $\mathbf{a}'\mathbf{y}_t$ is called a **long-run equilibrium relationship**: the $\mathrm{I}(1)$ series cannot drift arbitrarily far apart, because economic forces restrain them towards the equilibrium.

$\mathrm{I}(1)$ series that share a common stochastic trend can be combined into a stationary equilibrium error.

Economic examples. (i) The *permanent income hypothesis*: aggregate consumption is cointegrated with aggregate income. (ii) The *law of one price / no-arbitrage principle*: identical goods (or no-arbitrage-linked assets) should have stationary log price ratios if markets are integrated — useful, e.g., for antitrust market definition.

A statistical example. Let X_t be a random walk ($\mathrm{I}(1)$) and let $Y_t = \beta X_t + u_t$ with $u_t \sim \mathrm{I}(0)$. Then Y_t is also $\mathrm{I}(1)$, yet

$$Y_t - \beta X_t = u_t \sim \mathrm{I}(0),$$

so $\mathbf{a} = (1, -\beta)'$ is a cointegrating vector.

Remarks on cointegrating vectors

03

(i) **Non-uniqueness (scale invariance).** Cointegrating vectors are identified only up to scale: if $\mathbf{a}$ cointegrates $\mathbf{y}_t$, so does $\lambda\mathbf{a}$ for any $\lambda \neq 0$.

(ii) **Cointegrating rank.** For an N-dimensional $\mathrm{I}(1)$ system there may be r linearly independent cointegrating vectors, with $0 < r < N$; r is the **cointegrating rank**.

(iii) **Subspace.** Linear combinations of cointegrating vectors are themselves cointegrating vectors — they form a subspace.

Cointegrating vectors are fixed only up to scale, and there may be several of them.

What's special about cointegration? Superconsistency

04

OLS estimators of cointegrating vectors are **superconsistent** — they converge at the rate T rather than the usual $\sqrt{T}$. Remarkably, the parameters can be consistently estimated *despite* endogeneity (measurement error, simultaneity) that plagues typical macroeconometrics.
The intuition (Kennedy, 2008): a *wrong* parameter value produces an $\mathrm{I}(1)$ error term — infinite variance, an enormous sum of squared errors — whereas the *true* value produces an $\mathrm{I}(0)$ error with finite variance. Since OLS minimises the SSE, it homes in on the true value very quickly as T grows.

A wrong cointegrating vector leaves an $\mathrm{I}(1)$ residual — OLS punishes this so heavily that it converges at rate T.

Algebra — regressions among cointegrated processes with drift

Let $X_t = \alpha_X + X_{t-1} + \varepsilon_{1,t}$ and $Y_t = \beta X_t + u_t$, with u_t a zero-mean $\mathrm{I}(0)$ process independent of X_0 and $\varepsilon_{1,t}$. The OLS sampling error scaled by T is

$$T(\hat{\beta}_{OLS} - \beta) = \frac{T^{-2}\sum_{t=1}^{T} X_t u_t}{T^{-3}\sum_{t=1}^{T} X_t^2}.$$

The denominator converges to $\alpha_X^2/3 > 0$ (from the $O(T^3)$ growth of $\sum X_t^2$), while the numerator is $O_p(1)$. Hence $T^{-1}(\hat{\beta}_{OLS} - \beta) \to 0$, i.e. $\hat{\beta}_{OLS} - \beta = O_p(T^{-1})$ — convergence at rate T, not $\sqrt{T}$. This is superconsistency.

3 Testing for cointegration

The Engle–Granger two-step procedure

05

Step 1. Estimate the proposed cointegrating relationship by OLS,

$$Y_t = \alpha + \beta X_t + e_t,$$

where Y_t, X_t are both $\mathrm{I}(1)$; save the residuals $\hat{e}_t$.
Step 2. Apply a Dickey–Fuller test to the residuals, $\hat{e}_t = \delta_0 + \delta_1 \hat{e}_{t-1} + u_t$, testing $H_0 : \delta_1 = 0$ (a unit root in the residuals — **no cointegration**) against $H_1 : \delta_1 < 0$ (stationary residuals — **cointegration**). Rejecting H_0 is evidence in favour of cointegration.
Caveat. OLS chooses β to minimise the residual variance *even when the variables are not cointegrated*, so the residuals appear artificially stationary under the null. Using standard DF critical values would reject H_0 too often. One must therefore use **non-standard critical values** (more negative than the standard DF table) — see MacKinnon (1996).

Regress in levels, then unit-root-test the residuals — but with MacKinnon's stricter critical values.

Watch: the spurious regression problem

Watch: a cointegrated pair

Historical perspective *(not examinable)*

The non-stationary multivariate material closes with a historical arc — from the large-scale structural macro-models of Tinbergen (1939) through their predictive failures in the 1970s–80s, to the rise of time-series methods (ARMA, Box–Jenkins) and the VAR revolution (Sims, 1980). The thread running through it is the tension between economic theory and the statistical behaviour of the data — exactly the tension that cointegration helps to resolve.

---

4 Review questions

1. Explain the spurious regression problem for (a) deterministic and (b) stochastic trends. To what does the OLS slope converge when regressing two independent random walks with drift on each other?
2. Define cointegration and the cointegrating vector. Why is the cointegrating vector identified only up to scale?
3. What is superconsistency, and why does it arise for OLS estimators of a cointegrating relationship?
4. Describe the Engle–Granger two-step test for cointegration. Why must non-standard (MacKinnon) critical values be used in the second step?

---

5 Further reading

- **Granger & Newbold (1974)**, *Spurious Regressions in Econometrics*, Journal of Econometrics.
- **Engle & Granger (1987)**, *Co-integration and Error Correction*, Econometrica.
- **MacKinnon (1996)**, *Numerical Distribution Functions for Unit Root and Cointegration Tests*, JAE.
- **Dougherty (2011)**, *Introduction to Econometrics*, 4th edn. (historical context).
